Koa Shen

B.S. Candidate in Mechanical Engineering | UC Santa Barbara

koashensf@gmail.com (415) 810-8344 koashen.com San Francisco Bay Area / Santa Barbara, CA

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Marvell Hardware Engineering Hiring Team

Re: Summer 2027 Hardware Engineering Internship

Dear Marvell Hardware Engineering Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, and I am applying for a Summer 2027 hardware engineering internship at Marvell. I am interested in the engineering that makes data infrastructure and connectivity products work as complete, reliable systems. My background is in mechanical and electromechanical hardware rather than semiconductor design; I would bring experience in packaging, simulation, manufacturing, and hardware validation, along with a strong interest in learning from teams building complex connectivity technology.

At SLAC National Accelerator Laboratory, I built a Python/Drake digital twin for a beamline assembly with 30 degrees of freedom and 305 bodies. Millisecond collision checks revealed at least five physical pinch points and informed hardware redesigns. I created 15 reusable motion-stage models that are being adopted across three assemblies, and redesigned optics packaging, mounts, and detector strain relief. Combining simulation with load checks, CAD, and GD&T documentation taught me to connect model assumptions to physical constraints and make designs easier to manufacture and inspect.

My personal actuator project has given me direct experience integrating mechanical hardware with electronics and firmware. I designed and fabricated a compact 15:1 cycloidal reducer for a NEMA 17 stepper, then wrote PlatformIO C++ firmware for a TMC2209 driver and AS5600 encoder. On a single-joint bench setup, I validated closed-loop position correction and missed-step detection. With Gaucho Racing, I have also machined 33 suspension components from 4130 steel and 7075-T6 aluminum, holding critical bearing fits within 0.0005 inch, and led a 15-person team through safety-system fabrication and inspection.

I would bring Marvell a hands-on engineer who is comfortable crossing mechanical, software, and manufacturing boundaries, documenting decisions, and checking that designs behave as intended. I am excited to apply that foundation to hardware development and validation while growing my understanding of the systems behind high-performance connectivity.

Thank you for your consideration.

Sincerely,
Koa Shen